

6. [Maximum mark: 5]

A farmer grows two types of apples—cooking apples and eating apples. The weights of the apples, in grams, can be modelled as normal distributions with the following parameters.

Apple type	Mean μ	Standard deviation σ
Eating	100 g	20 g
Cooking	140 g	40 g

For each type of apple you can assume that 95 % of the weights are within two standard deviations of the mean.

- (a) Find the percentage of eating apples that have a weight greater than 140 g. [1]

The farmer grows a large number of apples of which 80 % are eating apples.

Both types of apples are picked and randomly mixed together in a cleaning machine.

After cleaning, the machine separates out those that have a weight greater than 140 g into a container.

- (b) An apple is randomly selected from this container. Find the probability that it is an eating apple. Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$. [4]

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